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Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-8

Lab Title: AI-ML Programs

Objective:

1. To learn and apply data pre-processing techniques that are fundamental for preparing datasets for AI and machine learning models.
2. To practice loading data from CSV files and handling missing values, which are common tasks in data cleaning.
3. To gain experience in encoding categorical variables, particularly converting text data into numerical formats suitable for machine learning algorithms.
4. To understand the importance of splitting data into training and test sets, a crucial step for model evaluation and generalization.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

Data Pre-processing in Machine Learning: Pre-processing is a critical step in the machine learning pipeline. It involves preparing raw data for modeling by cleaning, transforming, and structuring it. Proper pre-processing improves the quality of the input data, leading to better model performance and reliability.

- **Loading Data from a CSV File:**
  - **CSV (Comma-Separated Values) files** are commonly used for storing tabular data. They can be easily loaded into Python using libraries such as `pandas`, which provides the `read_csv()` function. This function reads the CSV file into a DataFrame, a structured data format that allows for efficient data manipulation and analysis.
- **Handling Missing Values:**
  - Missing values in datasets can lead to biased or incorrect models if not handled properly. Methods for handling missing values include:
    - **Finding missing values:** This can be done using functions like `isnull()` or `isna()` combined with `sum()` to count missing values in each column.
    - **Filling missing values:** Missing values can be filled using techniques such as imputation with the mean, median, or mode, or using more advanced methods like interpolation or predictive modeling.

- **Dropping Columns:**
  - Dropping unnecessary or irrelevant columns can simplify the dataset and improve model performance. This can be done using the `drop()` function in pandas, which allows you to remove specified columns.
- **Encoding Categorical Variables:**
  - Machine learning models require numerical input, so categorical variables (e.g., 'male' and 'female') need to be converted into numerical formats. This can be done using encoding techniques like:
    - **Label Encoding:** Converts each category into a number (e.g., 'male' to 1 and 'female' to 0).
    - **One-Hot Encoding:** Converts categories into binary vectors, which is useful for non-ordinal categorical data.
- **Splitting Input and Output Columns:**
  - For supervised learning, it's important to separate the input features (independent variables) from the output (dependent variable or target). This helps in structuring the data correctly for modeling.
- **Splitting Data into Train-Test Sets:**
  - Splitting the dataset into training and test sets is essential for evaluating the model's performance on unseen data. Typically, the data is split into a larger training set and a smaller test set using methods like `train_test_split()` from the `sklearn.model_selection` module. This step ensures that the model is trained on one portion of the data and validated on another, helping assess its generalization capability.

List of Programs in Assignment-8

1) Apply different pre-processing steps written below for the Titanic dataset.
   a) Write the Code to load the csv file.
   b) Write the code to find missing values in each column.
   c) Write the code to drop columns from the data.
   d) Write the code to convert 'male' to 1 and 'female' to 0 5.
   e) Write the code to fill missing values in 'Age' column.
   f) Write the code to split the Input and Output columns.
   g) Write the code the data in train-test set.

(Dataset Link: https://github.com/datasciencedojo/datasets/blob/master/titanic.csv)